

Kannan Thushanthan

PhD Candidate

Northumbria University

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Professional Summary

Civil engineer and researcher specialising in sustainable construction materials, waste valorisation, and durability of structures, with over four years of higher education teaching experience across the UK and Sri Lanka.

Professional Highlights

- **Sustainable Construction Expertise** - PhD research (submission July 2026) on transforming sewage sludge waste into high-strength lightweight aggregates, directly addressing low-carbon construction materials, circular economy, and UN SDGs 9, 11, and 12. Industry-funded by Northumbrian Water Group.
- **Peer-Reviewed Research Output** - Multiple high ranked Q1 journal articles in *Construction and Building Materials*, *Engineering Failure Analysis*, *Structures*, and *Bioresource Technology Reports*. Focus areas span sustainable materials, durability, and structural rehabilitation.
- **Higher Education Teaching Experience (4+ years)** - Module delivery and laboratory teaching at Northumbria University (Land Surveying, Hydraulics, Structural Design), SLIIT (Structural Analysis, Structural Dynamics), University of Moratuwa, and ESOF/Kingston University London (Advanced Materials). FHEA application under review (July 2026); committed to enrolling on PGCAP.
- **External Research Funding & Industry Engagement** - Senior Research Assistant on £38k Innovate UK AKT project with Doby Verrolec (cold-formed steel sections); contributed to EPSRC and collaborative PhD proposals; industry-sponsored CFRP retrofit consultancy on a wastewater equalisation tank.
- **Curriculum & Programme Contribution** - Contributed to curriculum development and university accreditation activities at SLIIT; designed coursework and assessment for the Kingston University London top-up programme delivered through ESOF.
- **Research Supervision & Mentoring** - Supervised undergraduate and MSc research projects on natural fibre reinforced concrete and durability assessment; mentored degree apprenticeship students at Northumbria.
- **International Collaboration** - Active research partnerships with Aston University (UK) and PNG University of Technology; cross institutional publication record across UK, Sri Lanka, and Australia.
- **Industry Practice in Construction** - Three years' construction industry experience including site supervision with China State Construction Engineering Corporation, structural assessment and CFRP strengthening of a near collapse wastewater tank, and renovation/risk assessment of rural school buildings in Sri Lanka.
- **Technical Software Proficiency** - AutoCAD, Revit, Civil 3D, SAP2000, ABAQUS, ANSYS, MIDAS Gen, MATLAB, Python - covering the industry-standard CAD tools.
- **Academic Leadership & EDI** - Elected Postgraduate Research Representative (2024-2025) and Equality & Diversity Representative (2025-2026) at Northumbria; STEM Ambassador (UK); demonstrated commitment to inclusive education and student support.

Education Profile

PhD in Civil and Environmental Engineering

Northumbria University, Newcastle upon Tyne

(Oct 2023-Expected July 2026)

Thesis: *Formulation of high-strength lightweight aggregates using renewable sewage sludge*

Principal Supervisor: A/Prof. Muhammad Rahman

Funding: Northumbrian Water Group (industry-funded PhD scholarship)

MPhil in Civil Engineering

Department of Civil Engineering, University of Moratuwa

(Mar 2020 - Jan 2023)

Thesis: *Bond characteristics and deterioration mechanisms of CFRP-concrete interfaces in acidic environments*

Principal Supervisor: Prof. J.C.P.H. Gamage

Funding: Senate Research Council (SRC) Research Scholarship

BSc (Hons) in Civil Engineering

Department of Civil Engineering, University of Moratuwa

(Nov 2015 - Jan 2020)

Upper Second-Class Honours; Dean's List - First Class equivalent in Final Year (GPA > 3.80)

Funding: Government of Sri Lanka Merit Scholarship

Research and Teaching Experience

PhD Researcher - Sustainable Construction Materials

Northumbria University, Newcastle upon Tyne, UK

(2023-Present)

Industry funded by Northumbrian Water Group

Supervisor: A/Prof. Muhammad Rahman

- Developing high strength lightweight aggregates from sewage sludge and GGBS as a low carbon alternative to conventional aggregates, contributing to circular economy in construction
- Conducting advanced microstructural characterisation (SEM-EDX, XRD, Raman, XRF) to evaluate phase evolution, sintering behaviour, and material performance
- Performing mechanical, durability, and thermal performance testing aligned with construction material standards
- Conducting life cycle and economic feasibility assessment to evaluate environmental sustainability of the developed aggregates
- Authored a critical review (*Bioresource Technology Reports*, Q1) on thermal valorisation of sewage sludge for sustainable construction applications; under review at *Construction and Building Materials*
- Established international research collaborations with Aston University (UK) and PNG University of Technology

MPhil Researcher - Durability of Structural Rehabilitation Systems

University of Moratuwa, Sri Lanka

(Mar 2020 - Jan 2023)

Funded by Senate Research Council (SRC)

Supervisor: Prof. J.C.P.H. Gamage

- Investigated long-term degradation mechanisms of CFRP-concrete interfaces under acidic, alkaline, and aqueous exposure to inform durable retrofitting of concrete infrastructure
- Applied non-destructive testing (UPV, rebound hammer) and SEM analysis to assess corrosion progression and chemical degradation
- Developed Weibull-based reliability models and proposed safety factors for CFRP-strengthened structures in aggressive environments
- Outputs: Q1 journal publication (*Engineering Failure Analysis*; under review at *Advances in Structural Engineering*) and three peer-reviewed conference papers

Research Assistant - Innovate UK AKT Project 2024

Northumbria University, in partnership with Doby Verrolec (£38,000 project) (April 2024 - Nov 2024)

- Contributed to the development of high-yield cold-formed steel channel sections for the construction sector
- Performed numerical modelling (ABAQUS) and experimental validation of structural performance
- Co-authored two Q1 journal publications in *Structures* on web crippling design of cold-formed sections

Undergraduate Research Project - CFRP Strengthening of Concrete Structures

University of Moratuwa, Sri Lanka

(2018 - 2020)

Supervisor: Prof. J.C.P.H. Gamage

- Developed a finite element model (ABAQUS) of NSM-CFRP-strengthened reinforced concrete curved beams under out-of-plane bending, validated against experimental testing

- Achieved Grade A* (>85 marks); foundational work for subsequent MPhil research

Research and Teaching Experience

Graduate Teaching Assistant

Northumbria University, Newcastle upon Tyne, UK

(Jan 2024 - Present)

- Delivering laboratory and field teaching across three core civil engineering modules: *Land Surveying* (KB4023, Year 1), *Hydraulics* (KB5025, Year 2), and *Structural Design* (KB4028, Year 2)
- Supporting undergraduate learning through small-group instruction, formative feedback, and assessment of practical work
- Mentoring degree apprenticeship and MSc students on coursework and project assignments

Lecturer

Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka

(Feb 2023 - Sept 2023)

- Co-taught and assisted on multiple modules within the undergraduate degree programme
- Delivered undergraduate modules across multiple year groups: *Structural Analysis* (CE2011, CE2042 - Years 1 & 2) and *Structural Dynamics and High-Rise Buildings* (CE4021 - Year 4)
- Designed lecture materials, coursework, and assessments aligned with intended learning outcomes
- Contributed to curriculum development activities in support of departmental accreditation
- Co-supervised undergraduate research projects on the durability of fibre-reinforced concrete

Visiting Lecturer

ESOFT Metro Campus / Kingston University London top-up programme

(Jan 2022 - May 2023)

- *Delivered the Advanced Materials module (AM2011, Year 2) on a UK-validated transnational programme*
- *Designed coursework and assessment in line with Kingston University London quality assurance standards*

Graduate Teaching Assistant

University of Moratuwa, Sri Lanka

(Jan 2021 - Jan 2023)

- Conducted laboratory and tutorial sessions across five undergraduate civil engineering modules: *Mechanics of Materials* (ME1032, Year 1), *Concrete Mix Design* (CE1132, Year 2), *Geotechnical Engineering* (CE3132, Year 3), *Survey Camp* (CE3913, Year 4), and *Highway Engineering* (CE4042, Year 4)
- Supervised laboratory safety, delivered technique demonstrations, and assessed practical reports

Research Funding & Projects

Industry Funded PhD Scholarship - Sustainable Construction Materials

(Oct 2023 - 2026)

Industry-Funder: Northumbrian Water Group

Awarded a fully funded PhD studentship to develop low-carbon lightweight aggregates from sewage sludge. Project addresses circular economy and waste valorisation priorities for the UK water and construction sectors.

Research Assistant - Innovate UK AKT Project

(April 2024 - Nov 2024)

Industry-Funder: Innovate UK; Industry partner: Doby Verrolec

Project value: £38,000

Contributed to development and structural validation of high-yield cold-formed steel channel sections. Outputs include two Q1 journal publications in *Structures*.

Research Proposal Development

(2024 - Present)

Northumbria University

Conducted preliminary investigations and contributed to drafting of research proposals for EPSRC, KTP, and collaborative PhD funding calls, supporting national and international partnership-building in sustainable construction materials.

MPhil Research Scholarship

(2020 - 2023)

Durability of CFRP-Strengthened Structures

Funder: Senate Research Council (SRC), University of Moratuwa

Awarded competitive research scholarship for postgraduate study on long-term performance of CFRP-strengthened concrete in aggressive environments.

Publications

Peer-Reviewed Journal Articles

Thushanthan, K., Mannan, M. A., Rahman, M., Poologanathan, K., & Rahman, M. (2026). Thermal valorisation of sewage sludge into artificial aggregates: A critical review of processes, environmental performance, and circular bioeconomy implications. *Bioresource Technology Reports*, 34(6), 102658. <https://doi.org/10.1016/j.biteb.2026.102658> (IF 4.3 - Q1; SDGs 9, 11, 12)

Rajanayagam, H., Upasiri, I., Poologanathan, K., Sifan, M., Rajadurai, A., Sherlock, P., **Kannan, T.**, & Kajienthan, K. (2026). Fire performance of LSF wall panels exposed to fire on both surfaces: A numerical approach. *Journal of Structural Fire Engineering* (ahead-of-print). <https://doi.org/10.1108/JSFE-07-2025-0035> (Q2).

Thirunavukkarasu, K., **Thushanthan, K.**, Poologanathan, K., Hajirasouliha, I., Rahman, M., Mohamed, S., & Higgins, C. (2026). Web crippling design rules for sigma sections under end one flange load case: A numerical study. *Structures*, 87(9), 111638. <https://doi.org/10.1016/j.istruc.2026.111638> (IF 4.3 - Q1).

Gray, D. T., Simwanda, L., Sifan, M., Poologanathan, K., & **Kannan, T.** (2026). Machine learning prediction of shear strength in cold-formed steel modular construction-optimised (MCO) beam. *Buildings*, 16(8), 1497. <https://doi.org/10.3390/buildings16081497> (IF 3.1 - Q1).

Sifan, M., Smith, K., Poologanathan, K., & **Kannan, T.** (2026). Finite element study of lightweight-concrete-filled hollow-flanged cold-formed steel beams under bending-shear interaction. *Buildings*, 16(7), 1370. <https://doi.org/10.3390/buildings16071370> (IF 3.1 - Q1)

Thushanthan, K., & Gamage, K. (2025). Durability assessment and reliability modelling of CFRP composites under simulated long-term exposure to acidic, alkaline, and aqueous environments. *Engineering Failure Analysis*, 110275. <https://doi.org/10.1016/j.engfailanal.2025.110275> (IF 5.7 - Q1; SDG 11).

Thirunavukkarasu, K., **Thushanthan, K.**, Poologanathan, K., Gunalan, S., Kanthasamy, E., Sifan, M., & Higgins, C. (2025). Web crippling design of cold-formed sigma sections - IOF load case. *Structures*, 81, 110326. <https://doi.org/10.1016/j.istruc.2025.110326> (IF 4.3 - Q1)

Justin, S., **Thushanthan, K.**, & Tharmarajah, G. (2025). Durability and mechanical performance of glass and natural fibre-reinforced concrete in acidic environments. *Construction and Building Materials*, 465, 140262. <https://doi.org/10.1016/j.conbuildmat.2025.140262> (IF 8.0 - Q1; SDGs 9, 11)

Manuscripts Under Review

Thushanthan, K., Mannan, M. A., Rahman, M., Poologanathan, K., & Rahman, M. *Sustainable utilisation of sewage sludge and GGBS for sintered lightweight aggregate production: compositional optimisation and microstructure, mechanical properties and performance evaluation*. Submitted to *Construction and Building Materials*.

Thushanthan, K., & Gamage, K. *CFRP-based rehabilitation of a joint-failed reinforced concrete wastewater tank under aggressive exposure conditions: materials performance and structural implications*. Submitted to *Advances in Structural Engineering*.

Peer-Reviewed Conference Papers

Thushanthan, K., Aponsu, G. M. D. N., & Gamage, J. C. P. H. (2022). Experimental study on tensile strength degradation of carbon fibre reinforced polymer (CFRP) composite in alkaline environment. *2022 Moratuwa Engineering Research Conference (MERCon)*, 1-6. <https://doi.org/10.1109/MERCon55799.2022.9906247>

Thushanthan, K., Alwis, P., & Perera, P. (2023). Investigation on integrity progression of rubberised concrete during the standard curing period. *Society of Structural Engineers Sri Lanka (SSESL) Annual Conference 2023*, 45-56.

Thirunavukkarasu, K., **Thushanthan, K.,** Poologanathan, K., Kanthasamy, E., Rahman, M., & Higgins, C. (2024). Web crippling behaviour of aluminium sigma sections: One flange load cases. *International Conference on Engineering, University of Jaffna (ICE 2024)*, Sri Lanka.

[Google Scholar Profile](#)

Consultancy & Industry Experience

Consulting Engineer

Wastewater Treatment Plant Rehabilitation

(Jan 2022 - Jan 2023)

UNIC Consultancy, University of Moratuwa, Sri Lanka

- Led the structural assessment and rehabilitation design of a near-collapse equalisation tank at an industrial wastewater treatment plant
- Conducted in-situ non-destructive testing, crack surveys, and microstructural analysis to evaluate degradation
- Specified and supervised CFRP-based strengthening interventions, extending service life and avoiding demolition - a directly sustainable rehabilitation outcome
- Outputs include a Q1 journal manuscript currently under review at *Advances in Structural Engineering*

Volunteer Renovation Engineer

School Buildings Risk Assessment

(Apr 2022 - Jun 2022)

UNIC Consultancy, University of Moratuwa, Sri Lanka

- Conducted structural risk assessment and renovation planning for three rural school buildings in Batticaloa and Ampara districts, Sri Lanka
- Carried out damage surveys, in-situ material testing, and structural evaluation
- Recommended cost-effective renovation measures appropriate for resource-constrained settings

Consultancy Team Member

Industrial and Educational Building Projects

(2022 - 2023)

- Contributed to multiple consultancy assignments including the Seethawaka Board of Investors (BOI) export processing zone renovation and a deteriorated three-storey building investigation at Thurston College, Colombo
- Activities included damage evaluation, in-situ testing, and renovation procedure recommendations

Trainee Civil Engineer

High-Rise Residential Construction

(Aug 2018 - Mar 2019)

China State Construction Engineering Corporation (CSCEC)

Sri Lanka Project: AVIC Astoria Luxury Residential Apartments, Colombo

- Supervised structural and finishing works on a high-rise residential project, including formwork inspection, concrete placement, waterproofing, and progress monitoring
- Gained practical experience in construction sequencing, quality control, and contractor coordination

Research Supervision & Mentoring

Co-mentor

Undergraduate Research Projects on Sewage Sludge Utilisation in Construction (2024 - Present)
Northumbria University, Newcastle upon Tyne, UK

- Mentoring undergraduate final-year students on research projects investigating sewage sludge utilisation in construction materials, working alongside the principal supervisor
- Trained students in advanced laboratory techniques including SEM imaging, compressive strength testing, and microstructural analysis
- Guided experimental design, data interpretation, and technical report writing aligned with research-led teaching practice

Assistant Supervisor

Undergraduate Research Projects (Feb 2023 - Sept 2023)
SLIIT, Sri Lanka

- Supervised undergraduate final-year research on the durability of glass and natural fibre-reinforced concrete in aggressive environments
- Guided experimental programme, durability testing, and statistical analysis
- Outputs: co-authored Q1 publication in *Construction and Building Materials* (IF 8.0)

Mentor - MSc and Degree Apprenticeship Students

Northumbria University, Newcastle upon Tyne, UK (2024 - Present)

- Mentoring MSc and degree apprenticeship students on coursework, project design, and laboratory technique
- Supporting students in laboratory safety, experimental planning, and report writing across structural and materials modules

Group Leader - Undergraduate Comprehensive Design Project

University of Moratuwa, Sri Lanka (2018 - 2020)

- Led a multi-disciplinary student team on a final-year capstone project (high-altitude sports training complex, Nuwara Eliya) covering ground improvement, foundation design, structural design, building services, and cost analysis
- Project outcome: Grade A* (above 85 marks)

Research Skills & Technical Proficiency

Materials Characterisation

- Scanning Electron Microscopy with Energy Dispersive X-ray Spectroscopy (SEM-EDX) - microstructural and elemental analysis
- X-ray Diffraction (XRD) - phase identification and crystallinity analysis
- X-ray Fluorescence (XRF) - chemical composition analysis
- Raman spectroscopy - molecular structure characterisation
- Thermogravimetric analysis (TGA)

Mechanical & Durability Testing

- Compressive, tensile, and flexural testing using Universal Testing Machines (UTM)
- Non-destructive testing - Ultrasonic Pulse Velocity (UPV), rebound hammer, crack mapping
- Accelerated ageing and durability testing in aggressive environments (acidic, alkaline, marine, sulphate exposure)
- Concrete mix design, casting, and curing for laboratory and field-scale specimens

Computational & Numerical Analysis

- Finite Element Analysis: ABAQUS, ANSYS (nonlinear modelling, contact mechanics, thermal analysis)
- Structural analysis and design: SAP2000, MIDAS Gen

- Statistical and reliability modelling: Weibull analysis, regression, life-cycle assessment principles

Software Proficiency

- CAD and BIM: AutoCAD (2D/3D), Revit, Civil 3D
- Programming: MATLAB, Python (data analysis, basic machine learning)
- Data analysis & visualisation: Origin, Microsoft Excel (advanced), Inkscape (scientific illustration)
- Project management: Microsoft Project
- Geospatial: ArcGIS

Sustainability & Performance Assessment

- Life-cycle assessment (LCA) principles for construction materials
- Environmental performance evaluation of waste-derived aggregates
- Microstructure-performance relationships in cementitious systems

Professional Membership & Teaching Fellowships

Higher Education Teaching

- Fellow of the Higher Education Academy (FHEA) - Application under review with Advance HE; outcome expected July 2026
- Committed to enrolling on the University's Postgraduate Certificate in Academic Practice (PGCAP) within 12 months of appointment

Professional Engineering Bodies

- STEM Ambassador - STEM Learning UK
- Associate Member - Institution of Engineers, Sri Lanka (IESL)
- Associate Member - Advances in Science and Technology Institute, Sri Lanka (ASTIS)
- Student Member - Society of Structural Engineers, Sri Lanka (SSESL)

Leadership & Academic Services

Equality, Diversity & Inclusion Representative

Northumbria University Students' Union (Elected role) (2025- 2026)

- Elected by postgraduate research students to represent EDI interests across the Mechanical and Civil Engineering postgraduate community
- Contributing to inclusive policy review, student support initiatives, and engagement with University EDI structures
- Completed Equality and Inclusivity Representative training delivered by Northumbria University Students' Union (2024)

Postgraduate Research Representative

Mechanical and Civil Engineering, Northumbria University (Elected role) (2024-2025)

- Elected by peers to represent the postgraduate research community within the department
- Acted as the primary student voice in PGR meetings, communicating researcher needs to academic leadership
- Supported induction and integration of incoming PGR students

Candidate - Vice President (Welfare)

Northumbria University Students' Union Election (2026)

- Standing for elected office at Students' Union level on a welfare-focused platform
- Demonstrating commitment to broader student leadership beyond departmental representation

STEM Ambassador

STEM Learning UK (2024-Present)

- Volunteer engagement in public outreach, school visits, and STEM enrichment activities
- Promoting civil and environmental engineering careers to under-represented groups in STEM

Awards & Scholarships

Industry-Funded PhD Scholarship

(2023 – 2026)

Northumbrian Water Group, UK Competitive, fully funded doctoral scholarship awarded for research on sustainable construction materials derived from sewage sludge.

Senate Research Council (SRC) Research Scholarship

(2020 - 2022)

University of Moratuwa, Sri Lanka Competitive postgraduate research scholarship awarded for the MPhil programme in Civil Engineering.

Government of Sri Lanka Merit Scholarship

(2015 – 2020)

Ministry of Education, Sri Lanka Awarded for undergraduate study in the Faculty of Engineering, University of Moratuwa, on the basis of national merit ranking.

Dean's List

(2019-2020)

Faculty of Engineering, University of Moratuwa Awarded for academic distinction in the final year of BSc (Hons) Civil Engineering (semester GPA > 3.80).

Additional Training & Professional Development

Northumbria University, UK

- PGR researcher development training (2024) - research integrity, supervision dynamics, academic writing
- Equality and Inclusivity Representative training, Students' Union (2024)

Sri Lanka Institute of Information Technology, Sri Lanka

- Laboratory safety and first aid training (2023)

Referees

A/Prof. Muhammad Ekhlasur Rahman

Principal PhD Supervisor

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Newcastle upon Tyne, UK

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A/Prof. Keerthan Poologanathan

KTP Project Supervisor and Research Collaborator

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